

SHETH L.U.J. AND SIR M.V. COLLEGE

Aim:- Performing logistic regression using glm() (R).

```
Console Terminal Background Jobs
R - R4.5.2 - ~/ -
> credit_data <- read.csv("Credit_Card_Applications.csv")
> credit_data$class <- as.factor(credit_data$class)
> log_model <- glm(Class ~ A2 + A3 + A7 + A10 + A13,
+ data = credit_data,
+ family = binomial)
> summary(log_model)

Call:
glm(formula = Class ~ A2 + A3 + A7 + A10 + A13, family = binomial,
    data = credit_data)

Coefficients:
(Intercept) -1.346e+00  3.021e-01 -4.456  8.36e-06 ***
A2           1.524e-05  8.501e-03  0.002  0.9986
A3           3.687e-02  1.974e-02  1.868  0.0618
A7           2.068e-01  3.941e-02  5.249  1.53e-07 ***
A10          3.613e-01  4.266e-02  8.469  < 2e-16 ***
A13          -5.182e-04  5.616e-04 -0.923  0.3561
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 948.16  on 689  degrees of freedom
Residual deviance: 718.89  on 684  degrees of freedom
AIC: 730.89

Number of Fisher Scoring iterations: 6

> probabilities <- predict(log_model, type = "response")
> plot(credit_data$A2, credit_data$class,
+ main = "Logistic Regression Curve (A2 vs Class)",
+ xlab = "A2",
+ ylab = "Class (0/1)",
+ pch = 16)
> ord <- order(credit_data$A2)
> lines(credit_data$A2[ord], probabilities[ord], col = "red", lwd = 2)
```

```
Source
Console Terminal Background Jobs
R - R4.5.2 - ~/ -
+ family = binomial)
> summary(log_model)

Call:
glm(formula = Class ~ A2 + A3 + A7 + A10 + A13, family = binomial,
    data = credit_data)

Coefficients:
(Intercept) -1.346e+00  3.021e-01 -4.456  8.36e-06 ***
A2           1.524e-05  8.501e-03  0.002  0.9986
A3           3.687e-02  1.974e-02  1.868  0.0618
A7           2.068e-01  3.941e-02  5.249  1.53e-07 ***
A10          3.613e-01  4.266e-02  8.469  < 2e-16 ***
A13          -5.182e-04  5.616e-04 -0.923  0.3561
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 948.16  on 689  degrees of freedom
Residual deviance: 718.89  on 684  degrees of freedom
AIC: 730.89

Number of Fisher Scoring iterations: 6

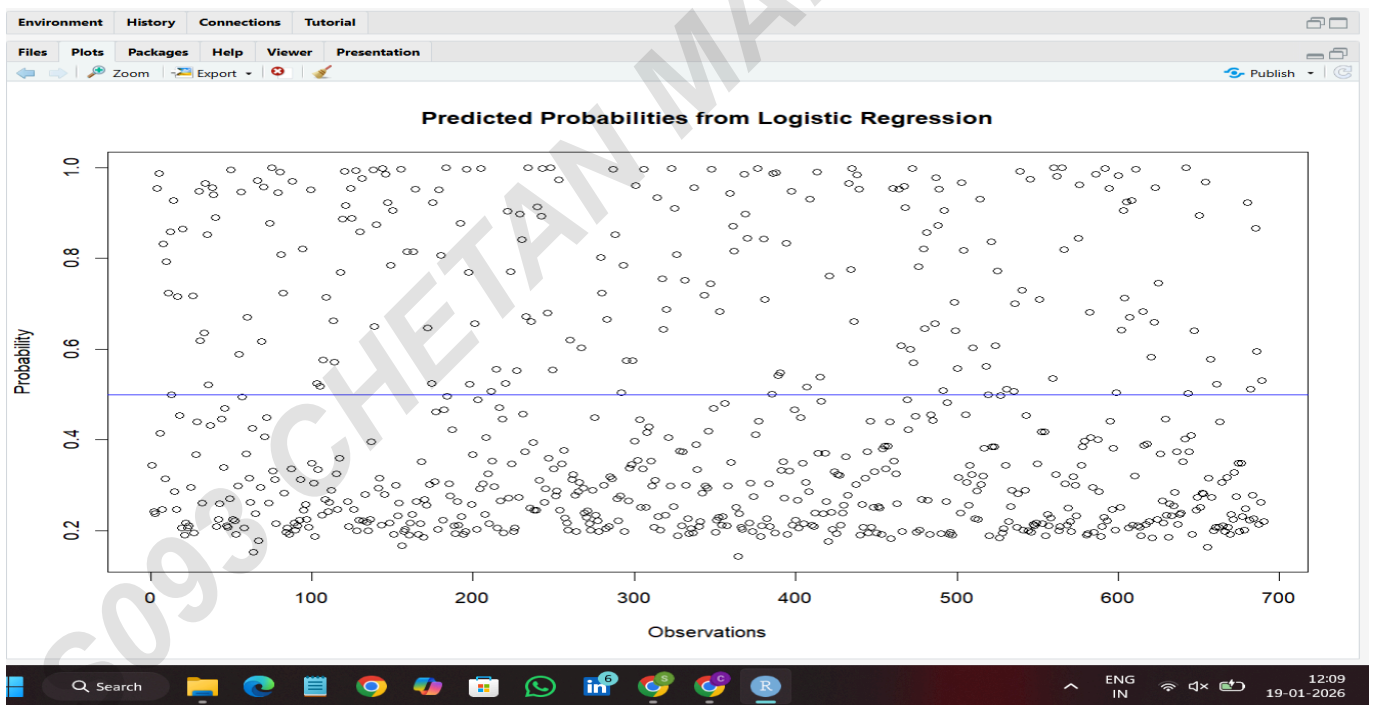
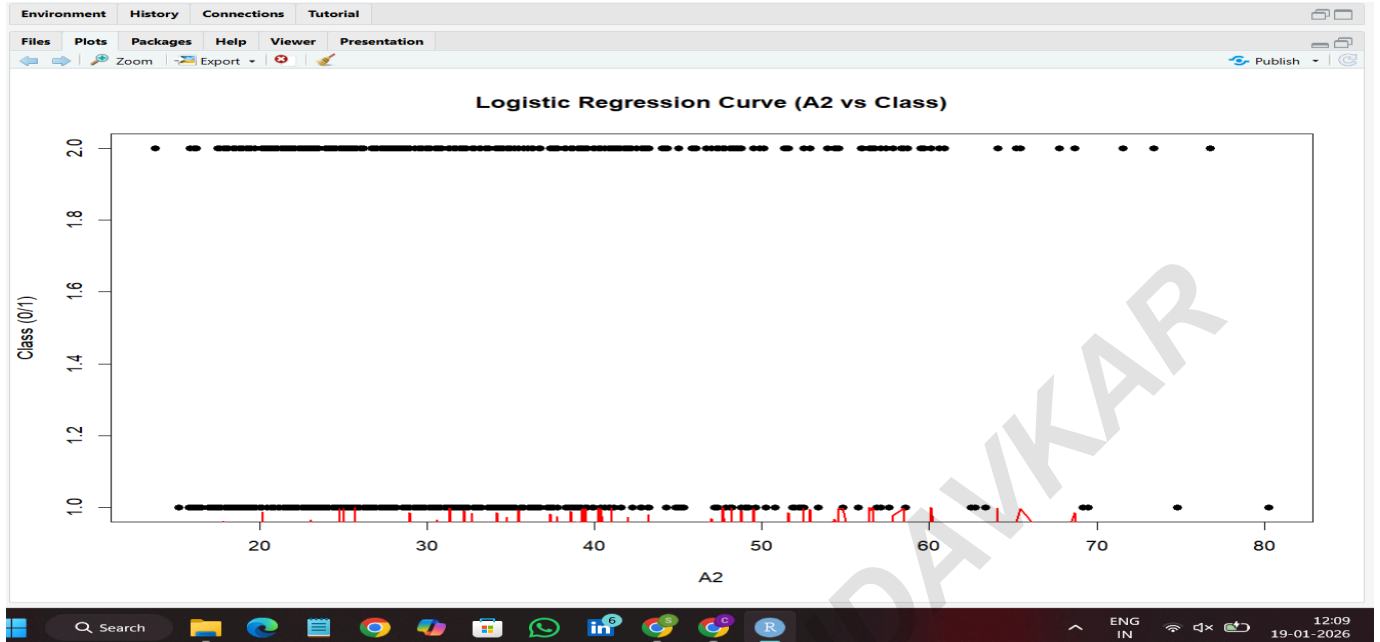
> probabilities <- predict(log_model, type = "response")
> plot(credit_data$A2, credit_data$class,
+ main = "Logistic Regression Curve (A2 vs Class)",
+ xlab = "A2",
+ ylab = "Class (0/1)",
+ pch = 16)
> ord <- order(credit_data$A2)
> lines(credit_data$A2[ord], probabilities[ord], col = "red", lwd = 2)
> plot(probabilities,
+ main = "Predicted Probabilities from Logistic Regression",
+ ylab = "Probability",
+ xlab = "Observations")
> abline(h = 0.5, col = "blue")
>
```

NAME:- CHETAN MANDAVKAR

ROLL NO. S093

SUBJECT:- Data Analysis With SAS / SPSS / R

SHETH L.U.J. AND SIR M.V. COLLEGE



NAME:- CHETAN MANDAVKAR

ROLL NO. S093

SUBJECT:- Data Analysis With SAS / SPSS / R

SHETH L.U.J. AND SIR M.V. COLLEGE

S093 CHETAN MANDAVKAR

NAME:- CHETAN MANDAVKAR

ROLL NO. S093

SUBJECT:- Data Analysis With SAS / SPSS / R